Course Title: Technology, Environment and Society

HUM 4147

Content

Air Pollution

Air Pollution



Air Pollution

5 Major Pollutants:

1.) Carbon Monoxide
2.) Sulfur Dioxide
3.) Nitrogen Dioxide
4.) Particulate Matter
5.) Ground Level Ozone

Air Pollution

Carbon Monoxide

- colorless, odorless gas
- produced when carbon does not burn in fossil fuels
- present in car exhaust
- deprives body of O2 causing headaches, fatigue, and impaired vision





Air Pollution

Sulfur Dioxide

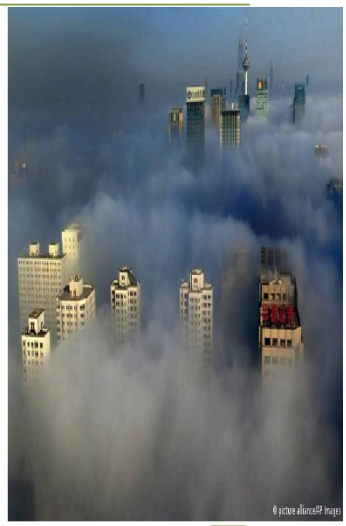
- produced when coal and fuel oil are burned
- present in power plant exhaust
- narrows the airway, causing wheezing and shortness of breath, especially in those with asthma



Air Pollution

SMOG

- Combination of gases with water vapor and dust
  - Combination of smoke and fog
    - Forms when heat and sunlight react (photochemical smog)
      - Occurs often with heavy traffic, high temperatures, and calm winds



Air Pollution

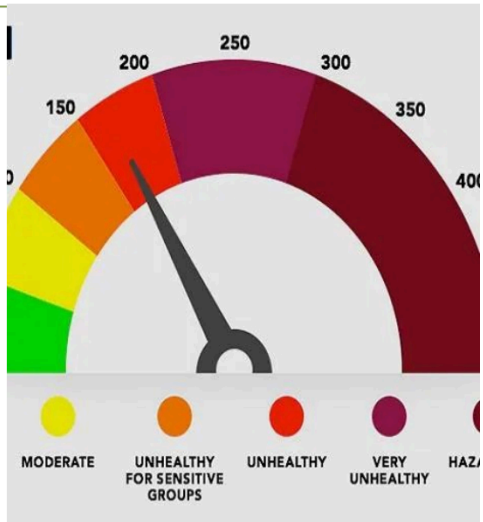
Other Problems…

- ✓ Green House Effect
- ✓Global Warming
- ✓Thinning of Upper Ozone Layer
- ✓Acid Rain
- ✓Indoor Air Pollution

Air Pollution

AQI: Air Quality Index

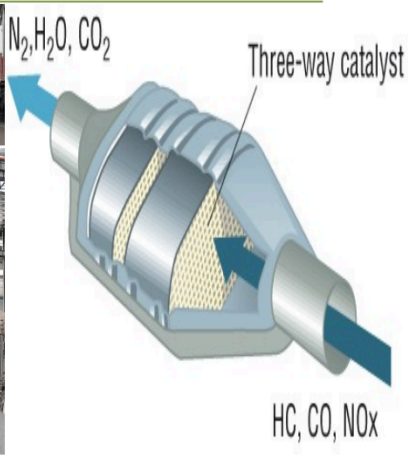
- The Air Quality Index (AQI) is **used for reporting daily air quality**.
- Indicates pollutant level in the air.
- It tells how clean or polluted air is, and what associated health effects might be a concern.
- The AQI focuses on health effects you may experience within a few hours or days after breathing polluted air.
- Ranges from 0 (least concern) to 500 (greatest concern)



Air Pollution



Mechanical filter https://www.youtube.com/watch?v=6IJt2Z0pQDk



Catalytic Converter

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Mechanical filter

Mechanical filter

A mechanical filter is a type of filter that is primarily designed to remove suspended solid particles. It can be used to maintain the cleanliness of a system, protecting it from abrasive wear or corrosion. Mechanical filters can be made of a fine mesh of synthetic polymer fibers.
A mechanical filter can be extremely useful in industrial settings such as in cooling systems where the environment can be extremely corrosive.
They are used for air-purifying respirators that mechanically stops particulates from reaching the wearer's nose and mouth. They come in multiple physical forms. With a properly functioning mechanical filter, industrial problems such as contamination and faults can be avoided, making mechanical filters an important part of industrial operations. With a clean and operational system, problems with corrosion can be prevented.



Mechanical filters remove contaminants from air in the following ways:-
1. by *interception* when particles following a line of flow in the airstream come within one radius of a fiber and adhere to it;
2. by *impaction*, when larger particles unable to follow the curving contours of the airstream are forced to embed in one of the fibers directly; this increases with diminishing fiber separation and higher air flow velocity
3. by an enhancing mechanism called *diffusion*, where gas molecules collide with the smallest particles, especially those below 100 nm in diameter, which are thereby impeded and delayed in their path through the filter; this effect is similar to Brownian motion and increases the probability that particles will be stopped by either of the two mechanisms above; it becomes dominant at lower air flow velocities.